

1.

The function f is defined by $f(x) = 3^x$. The function g is an increasing linear function. In the xy -plane, the graphs of $y = f(x)$ and $y = g(x)$ intersect at two points (a, j) and (b, k) , where $j < k$. When $g(x) > f(x)$, which of the following must be true?

- A) $x > k$
- B) $x < j$
- C) $x < j$ or $x > b$
- D) $a < x < b$

2.

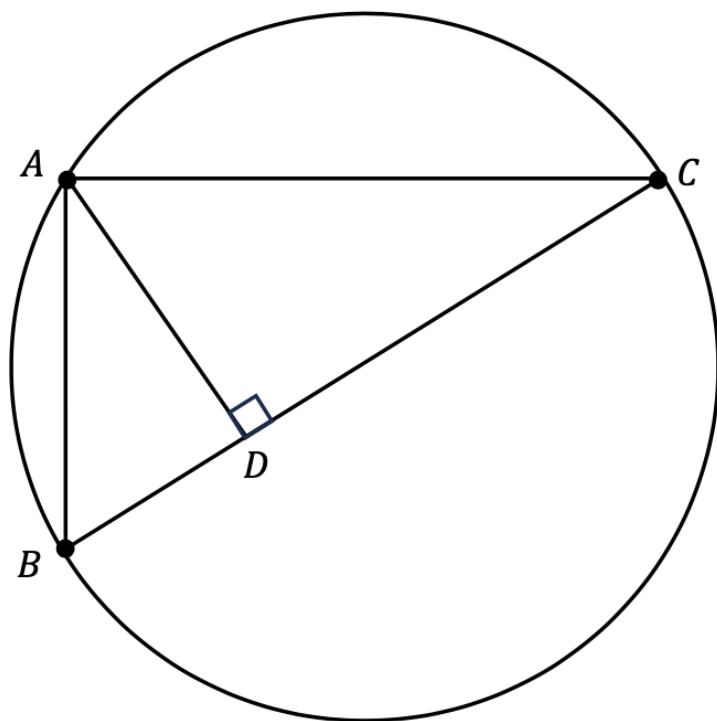
$$\begin{aligned}7x - 9y &= 39 \\ 35x - 45y &= 195\end{aligned}$$

For any real number r , which of the following points lies on the graph of both equations?

- A) $(3r, \frac{7r-13}{3})$
- B) $(\frac{7r+13}{3}, 3r)$
- C) $(-\frac{7r}{-9} + 39, \frac{7r}{9} + 195)$
- D) $(\frac{r}{5} + 39, -\frac{r}{5} + 195)$

3.

In the given figure, BC is the diameter of the circle. If the length of BC is equal to 132 and the length of AB is equal to $\sqrt{363}$, what is the value of $\frac{BC}{BD}$?



Note: Figure not drawn to scale.

4.

The given quadratic function

$54x^4 + 219x^2 + 105$ has factors in the form $(k)(ax^2 + b)(cx^2 + d)$. If a, b, c, d , and k are integers, what is the smallest possible value of ab ?

5. In the given equation, k is a positive constant.
The product of the solutions to the equation is 154. What is the value of k ?

$$\frac{3}{5}(5x + 11)(x + \sqrt{5k + 10})(x - \sqrt{5k + 10}) = 0$$